

Sabarinath Sri Dhandapani

Aerospace Engineer

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PROFILE

Aerospace engineer with an MSc from the University of Leicester, working across CFD, flight dynamics, control system design, and thermal analysis. Built a 7-state F-16 longitudinal flight dynamics model in MATLAB/Simulink and designed an LQR altitude-hold controller around it, verifying performance held up under a +20% perturbation in the aircraft dynamics before trusting the result. Separately, used ANSYS Fluent to estimate propulsion performance across multiple configurations and correlated simulation predictions against physical wind-tunnel test data. Writes things down properly — technical reports, documented assumptions, results that hold up when someone else checks them. Looking for an entry-level engineering role in CFD, flight dynamics, control systems, or thermal analysis within civil aerospace, motorsport, or related high-performance engineering sectors.

EDUCATION

MSc Aerospace Engineering

2024 – 2025

University of Leicester, UK

- Specialised in aerodynamic modelling, flight dynamics, computational simulation, and control systems engineering.
- Core modules: Advanced Fluid Dynamics (turbulence modelling theory, boundary layer theory, heat transfer and thermal analysis), Computational Methods, Systems Modelling, Flight Mechanics, Control Engineering.

BEng (Hons) Aeronautical Engineering

2019 – 2023

Kumaraguru College of Technology, India — GPA 3.2/4.0 (UK equivalent: Upper Second-Class Honours, 2:1)

- Core strengths: aerodynamics, structures, propulsion systems, CFD.
- Leadership: Treasurer — KI Aero Modelling Club; Event Lead — SEDS KCT.

TECHNICAL SKILLS

- Simulation & Control: MATLAB/Simulink, LQR control design, state-space modelling, dynamic and steady-state system analysis
- CFD & Thermal Analysis: ANSYS Fluent (MRF), STAR-CCM+/Simcenter, COMSOL — thrust estimation, aerodynamic loading, heat transfer, conjugate heat transfer, fluid-structure interaction, aeroelasticity
- CAD & Design: CATIA, SOLIDWORKS (CSWP Certified) — mechanical design, complex assembly modelling, parametric part design, structural and aerospace design modelling
- Systems Engineering: Requirements definition, verification & validation (V&V), performance analysis, data correlation
- Programming: Python, MATLAB, C — simulation, data processing, automation
- Embedded & Hardware: Arduino, Proteus, STK, OpenRocket
- Documentation: LaTeX, MS Office — technical reporting and presentation

PROJECT EXPERIENCE

Altitude-Hold Autopilot Design for a Fixed-Wing Fighter Aircraft Model

May 2025 – Sep 2025

University of Leicester — MSc Research Project

- Built a 7-state linear state-space model of F-16 longitudinal dynamics (altitude, pitch angle, forward velocity, angle of attack, pitch rate, plus two further stability states) in MATLAB/Simulink, with throttle and elevator as the two control inputs; augmented it with an integral altitude-error state (8 states total) and confirmed full controllability (rank 8) before designing the controller.
- Designed a full-state-feedback LQR controller with integral action, tuning the Q/R weighting matrices to trade off tracking speed against control effort.
- Achieved a 100 ft step response with 0.9% overshoot, 5.74 s settling time (2% band), and 0.023 ft steady-state error in the nominal case.
- Stress-tested the design against a +20% increase in the state matrix A: settling time extended to 7.22 s and overshoot fell to 0%, with steady-state error still just 0.019 ft — confirming the controller held stability and tracking accuracy outside the nominal design point.

LiDAR-RADAR Sensor Fusion for Moving Target Detection

Aug 2022 – Apr 2023

Kumaraguru College of Technology — Undergraduate Team Project (3-person team)

- Owned the Kalman filter design and MATLAB simulation workflow for the team, building a nonlinear moving-target motion model to fuse LiDAR and RADAR measurements into a single state estimate.
- Chose an Unscented Kalman Filter over the more conventional EKF after evaluating both against the model's nonlinear dynamics, avoiding the Jacobian linearisation error the EKF would introduce.

Dynamic Thrust Estimation of a Fixed-Wing UAV

Jan 2022 – Jun 2022

Kumaraguru College of Technology — Undergraduate Project

- Set up and solved a Moving Reference Frame (MRF) CFD model in ANSYS Fluent to estimate propeller thrust, defining rotating-zone boundaries and interface conditions rather than using a simplified actuator-disk approximation.
- Ran the model across multiple propulsion configurations and compared thrust-to-drag trends between them, identifying which configuration gave the best performance trade-off — output that fed directly into the team's propulsion R&D decision.

Metal-Detection Security Drone — Structural Design

Aug 2021 – Dec 2021

Kumaraguru College of Technology — Undergraduate Project

- Designed a tricopter airframe in CATIA around an integrated metal-detection sensor payload, applying lightweight aerospace structural principles to a non-standard multirotor configuration.
- Iterated the design to balance structural integrity, airframe weight, and payload placement — competing constraints that don't have a single dominant solution and had to be actively traded off.

RELEVANT EXPERIENCE

Intern Trainee

Feb 2022 – Apr 2022

SASTRA University, India

- Worked across composites manufacturing, 3D printing, and aerodynamic wind-tunnel testing, correlating simulation predictions with physical test data.

Intern Trainee

Jun 2022 – Sep 2022

CIPET, Chennai, India

- Analysed industrial manufacturing and quality-control processes, building practical understanding of production-floor engineering standards.

Design Intern

Oct 2020 – Jan 2021

Space Technology and Aeronautical Rocketry (STAR), India

- Designed and simulated a model rocket and static test bed using OpenRocket and Proteus, gaining early hands-on exposure to aerospace R&D workflows.

Logistics Associate

Sep 2026 – Present

GXO Logistics, UK — promoted twice within a year (Warehouse Operative → Warehouse Associate → Logistics Associate); full-time role held in parallel with an active job search in aerospace engineering, not a career direction.

CERTIFICATIONS

- Ansys Certified Associate — Foundations in Fluid Dynamics
- Ansys Certified Associate — Foundations in Heat Transfer
- COMSOL Multiphysics Training — Aerothermal & Conjugate Heat Transfer
- COMSOL Multiphysics Training — Fluid-Structure Interaction for Aerospace Structures
- MATLAB Onramp & Simulink Onramp — MathWorks
- STAR-CCM+ for a Beginner — Flowthermolab
- Dassault Systèmes Certified SOLIDWORKS Professional (CSWP) — Mechanical Design
- Microsoft Certified: Azure Fundamentals (AZ-900) — Microsoft

ADDITIONAL INFORMATION

- Right to work: UK Graduate visa — eligible to work full-time in the UK; sponsorship may be required for continued employment beyond the visa term.
- Notice period: 1 week.
- Languages: English (fluent), Tamil (native).
- Location: Leicester, UK — open to relocation across the UK and Europe.